

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)
Course Name: Applied Chemistry Course Code: CHM-111 Examination: 2nd Periodical
Scheme: CBCS Branch: CE/CSE Semester: 2nd
Question Paper for 2nd Periodical Examinations (April, 2024)

Time: Two hours

Max. Marks: 50

Note: All questions are compulsory.

Course Outcomes: *At the end of course students will be able to:*

1. Apply the concept of water treatment and electrochemistry in daily life and industrial applications.
2. Design various types of methods to prevent corrosion.
3. Apply the concept and principles of spectroscopic techniques in material characterization.
4. Evaluate various types of fuels on the basis of their properties and uses.
5. Summarize role of chemistry in ICT.
6. Critically examine the role of various nano-materials and polymers in engineering field.

Part A

Q1.

- (i) (CO1) Differentiate between molar and equivalent conductance.
- (ii) (CO1) Discuss the effect of hydrogen bonding on vibrational frequency.
- (iii) (CO3) What is Hooke's Law? Give its applications.
- (iv) (CO5) List out main chemicals (metal/metal oxides) used in touch enabled display devices?
- (v) (CO6) Discuss the role of nano-materials in our daily life.

(2×5=10)

Part B

- Q2. (CO1) (i) What is glass electrode? Discuss role of glass electrode in pH determination. **(5)**
(CO1) (ii) Explain working of Lithium ion batteries. **(5)**
- Q3. (CO3) Discuss in detail principle and applications of UV-Visible spectroscopy. **(10)**
- Q4. (CO6) Write short notes on followings: (a) Graphene, (b) Carbon nanotubes (c) Fullerenes **(10)**
- Q5. (CO5) (i) How chemistry and information & communication technology are related to each other. **(5)**
(CO6) (ii) What are conducting polymers? Explain mechanism of conduction in polyacetylene. **(5)**

Course Advisor: Dr. Ajnesh Singh